

A Review of Computational Models for Cognitive Distortion Classification

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Abstract—Irrational thought processes known as cognitive distortions are a key idea in cognitive behavioral therapy (CBT), which is used to treat disorders like anxiety and depression. The growth of digital mental health services has made the automatic classification of these aberrations from text a crucial goal for computational linguistics. This overview summarizes the field by following the methodological development of computational models from early feature-based systems to contemporary transformer architectures, as well as the psychological taxonomies that define the categorization labels.

I. INTRODUCTION

Over time, people may unconsciously reinforce habitual, harmful, and illogical thought habits known as cognitive distortions. These erroneous beliefs were first noted by psychiatrist Aaron Beck in the 1960s as part of his creation of Cognitive Behavioral Therapy (CBT), and they are now acknowledged as frequent causes of emotional suffering.

The CBT model asserts a direct connection between a person's cognitions (automatic thoughts), emotions, and behaviors. According to Beck's theory, the interpretation (cognition) of a circumstance is what generates distress rather than the reality itself. Therefore, negative emotions are reinforced by negative or incorrect perceptions, which can result in maladaptive behaviors and a self-perpetuating loop.

CBT is a two-step process: The first step is for the patient and therapist to learn to recognize these automatic, negative thoughts as they happen. The second step is for the patient to learn how to question, reframe, or change these thoughts so that they are more realistic and helpful.

Computational cognitive distortion classification as a whole can be viewed as an effort to automate and operationalize the identification phase of cognitive behavioral therapy. There are a lot of possible uses, from informing a human therapist about a patient's dominant thought patterns to giving users in a journaling app real-time feedback. [4]

II. CORE DISTORTION CATEGORIES - A WORKING TAXONOMY

According to Beck and Burns' research [1], the following categories are most commonly seen in computational studies, however lists can range from 10 to 15 or more:

- **All-or-Nothing Thinking:** Seeing things, people, or one-self in absolute, binary terms with no room for compromise. Example: "I've blown my diet completely" after consuming one spoonful of ice cream.
- **Overgeneralization:** Using phrases like "always" or "never" to describe a single unfavorable incident as a never-ending cycle of defeat.
- **Mental Filter:** Focusing just on one unfavorable detail, which "discolors a beaker of water," while disregarding all favorable aspects of a circumstance.
- **Jumping to Conclusions:** Making unfavorable assumptions about a situation without enough proof. Two different distortions often use this as a parent category:
- **Mind reading** is the act of assuming, without verifying, that someone is responding negatively to you. For instance: "My friend replied late to my text; they must be upset with me".
- **Catastrophizing:** Predicting a bad, frequently worst-case scenario.
- **Labeling:** An extreme type of overgeneralization in which a person or circumstance is given a negative, absolutist, and frequently emotionally charged label. For instance, believing "I am a failure" rather than "I made a mistake."
- **Discounting the Positive:** Declaring that good experiences "don't count" for whatever reason.
- **Emotional Reasoning:** "I feel it, therefore it must be true" is an example of this, which assumes that subjective emotions are objective facts.

III. FORMULATING THE CLASSIFICATION TASK

Three main formulations have emerged from the conversion of these ambiguous psychological notions into exact machine-learning problems, each with a unique trade-off between model performance and clinical validity.

A. Binary Classification (Detection)

A text is categorized as "Distorted" or "Not-Distorted" in this most basic version. The "important first step" in CBT, which is merely identifying the existence of distorted thinking, is reflected in this task. The best performance metrics are frequently obtained from this task. For instance, in this

TABLE I
EVOLUTION OF MODELS

Approach	Dataset	Performance
Classical ML	Small-to-medium labeled crowdsourced logs	F1 often low-to-moderate
Early Deep Learning	Crowdsourced logs with contextual metadata	F1 often in 50–70%
Transformer fine-tuning	Labeled datasets, class sets vary widely	Micro-F1 often reported in 65–80%

binary detection challenge, Shickel et al. (2020) [3] obtained a weighted F1-score of 0.88.

B. Multi-Class (Single-Label) Classification

The model is required to assign one distortion label from a predetermined list (such as 10, 14, or 15 classes) in this “N-ary text classification” task. As said, this formulation is therapeutically problematic despite being computationally simple. Because numerous distortions can co-occur with a single idea, it forces a false choice. Performance on this task is consistently worse; the same study by Shickel et al. (2020) [3] found a weighted F1 of only 0.68 for the 15-class task, despite achieving F1=0.88 on detection.

C. Multi-Label Classification

This formulation is the most computationally complex and therapeutically realistic. It enables the simultaneous assignment of several distortion categories to a single text. In reality, research frequently models this as “N” independent binary classifications, one for each label. Several cutting-edge research that seek to capture the subtleties of clinical literature employ this method.

Thus, the task formulation decision is a clear trade-off. Binary detection, the most straightforward job, has the lowest clinical utility but the greatest F1-score. The most challenging challenge for both human annotation and model training is the most complicated and clinically meaningful task (multi-label), which leads to lower reported metrics. This discrepancy contributes to the explanation of the field’s “fragmented” and uneven evaluation procedures.

IV. LITERATURE REVIEW

The technical approaches used to classify cognitive distortions have evolved from feature-based models to context-aware transformers, reflecting the larger development of the NLP discipline [1]

A. Conventional Feature-Based Machine Learning

Traditional classifiers that rely on manually designed characteristics were used in early and, in some circumstances, still competitive experiments.

- **Shickel et al. (2020)** [3] reported that their Linear Regression (LR) model with Term Frequency-Inverse Document Frequency (TF-IDF) with F1=0.68 performed better than their own Convolutional Neural Network (CNN) models.
- **Shreevastava and Foltz (2021)** [7] used a Support Vector Machine (SVM) to attain a strong F1-score of 0.77 in a detection challenge.

B. Early Deep Learning Models (Static Embeddings)

In order to automate feature extraction, the field quickly switched to deep learning.

- **Rojas-Barahona et al. (2018)** [6] employed a Convolutional Neural Network (CNN) with Global Vectors for Word Representation (GloVe) embeddings for multi-label classification on a sizable dataset. Their research laid the groundwork for showing that deep learning models may “significantly outperform” non-deep learning models in this particular challenge.
- **Mostafa et al. (2021)** [2] pointed out a critical issue with the above research, they employed Long Short-Term Memory (LSTM) with GloVe: these models were prone to overfitting when trained on the small and artificial datasets that were accessible to the public.

C. Transformer-Based Architectures (Contextual Embeddings)

Large, pre-trained transformer models with contextual embeddings define the present state-of-the-art.

- **Tauscher et al. (2022)** [10] - A tuned Bidirectional Encoder Representations from Transformers (BERT) model obtained an F1-score of 0.62 for binary detection in clinical texts
- **El Bolock et al. (2024)** [8] discovered that a hybrid CNN architecture using pre-trained Sentence-BERT (S-BERT) embeddings was their best-performing model on a sizable crowdsourced dataset, achieving a high F1-score of 0.84.
- **Qi et al. (2024)** [9] discovered that LLMs fall well short of supervised models in the absence of fine-tuning (e.g., by 31.53 percentage points on F1).

The growth of one essential capability—the modeling of context—from SVMs to LSTMs to BERT is evident. Ambiguity is the main obstacle in this task. Only in certain contexts—like “I should be perfect”—is a “should” statement distorted; in other contexts—like “You should turn left”—it is not. It is difficult for a feature-based model to distinguish between the two. Local, sequential context is modeled by an LSTM. A transformer is far more skilled at this disambiguation since it models deep, bidirectional context.

The data, however, indicates that “newer is not always better.” A key synthesis is indicated by the results that an LR performed better than a CNN, that LSTMs overfit on poor data, that a hybrid CNN+S-BERT was a top performer, and that refined models continued to outperform zero-shot LLMs [5]: the field is data-limited, not model-limited. The quantity, quality, and contextual richness of the training data seem to be more significant than the model architecture selection.

V. DATASETS, BENCHMARKS, AND PERFORMANCE

The datasets used for training and evaluation are intrinsically tied to the empirical evidence for model performance. Direct model comparison is challenging due to the field's reliance on inconsistent, proprietary, or artificial data.

A. Data Sources and Annotation Challenges

The absence of sizable, publicly available, high-quality, annotated corpora is a major obstacle [1]. Usually, data comes from three different domains:

- **Crowdsourced/Synthetic:** Many researchers use crowdsourcing to generate their own datasets in order to address the lack of available data. Large datasets can result from this (El Bolock et al. (2024) gathered 34,370 items, for example), although these instances do not have the ambiguity and clinical realism of true skewed thinking. [1]
- **Social media:** Data is scraped by researchers from Chinese social media sites such as Reddit and Tumblr. Although this data is "in-the-wild" and natural, there are serious ethical issues with user consent and privacy [1], as well as a high degree of annotation difficulty.
- **Clinical/Therapeutic:** The most useful and accurate data, however it is "often inaccessible due to privacy constraints" [1]. Examples include data from an online therapy program, the 7,354 clinical text messages from patients with severe mental illness utilized by Tauscher et al. (2022), and the private 500k-post corpus from a peer-to-peer therapy platform.

VI. METHODOLOGY

A. Pre-trained Transformer Approach

We developed a hybrid approach that uses (1) rule-based pattern matching for cognitive distortions, (2) a pretrained transformer for fine-grained emotion classification, and (3) lightweight sentiment lexicons - Valence Aware Dictionary and sEntiment Reasoner (VADER) and TextBlob:

- We generated a distribution of emotion labels and a principal emotion using an open-source, pretrained transformer emotion classifier (via the Hugging Face pipeline).
- **VADER** polarity scores to measure coarse positive, neutral, and negative sentiment appropriate for brief, social-style writing, and
- **TextBlob** polarity/subjectivity to offer a different viewpoint on lexical sentiment.

The analyzer combines these three signals to report a primary emotion as well as a "overall sentiment" based on the compound score of VADER.

The main method for detecting cognitive distortion is rule-based. For frequent distortion categories, such as all-or-nothing, catastrophizing, customization, should-statements, and labeling, we developed a set of regular expressions. The detector identifies patterns that match, provides example matches and counts for each category, and calculates a distortion score that is normalized

B. Synthetic Neural Network Approach

Instead of using big, pretrained transformer models, we created a fully customized training pipeline to provide a useful and independent cognitive distortion classification system. Our method works well for experimental studies.

- **Synthetic Dataset Construction** - First, we identified ten typical kinds of cognitive distortion, such as labeling, personalization, catastrophizing, and all-or-nothing thinking. We selected sample seed examples for each category. After that, we used controlled text augmentation to significantly increase the dataset by adding noise modifiers (such "really" and "extremely") and creating multiple variations for each class.
- **Text Vectorization** - We employed TF-IDF vectorization (unigrams to trigrams, 500-feature limit) in place of contextual embeddings. This decision offers excellent performance on fundamentally repeating statements, such as distortions, while maintaining the system's lightweight and interpretability.
- **Neural Network Architecture** - We trained two separate feed-forward neural networks - Binary Distortion Detector (Dimensions of input: 500, Layers hidden: 256 $\rightarrow$ 64 $\rightarrow$ 32, One neuron with a sigmoid output) and Emotions Classifier (Output: quantity of emotion classes). In order to prevent overfitting, both models employ dropout regularization and ReLU activations, which helped maintain performance even using a fake dataset.
- **Pattern-based detection** - We used regular expressions to create a rule-based pattern matcher in addition to the neural classifier. This hybrid strategy enables identification of particular categories that cannot be distinguished by binary classification alone

VII. RESULTS

A. Pre-trained Transformer

- For Distortion Detection: Accuracy, Precision, Recall and F1-score are 90%
- For Sentiment Detection, Accuracy is 80%

The analyzer returns Distorted Score, Cognitive Distortions Detected, Emotional Analysis, Mental Health Assessment

B. Synthetic Neural Network

The Adam optimizer was used to train both models for 50 epochs at a learning rate of 0.001.

The Emotion Classification Model has the following results

- Accuracy: 0.51
- Joy: High precision (0.74), high recall (0.93), strong F1-score (0.83)
- Sadness: Good precision (0.68), good recall (0.81), good F1-score (0.74)
- Anger: Low precision (0.32), but high recall (0.83).

The analyzer returns Distortion Score, Specific Distortions Detected (with occurrences), Primary Emotion (with percentage, Concern Level

Fig. 1. User inputs journal entry

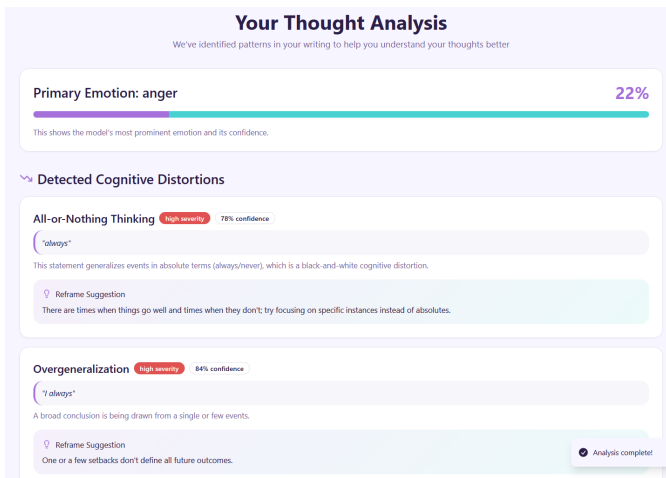


Fig. 2. Analysis Results

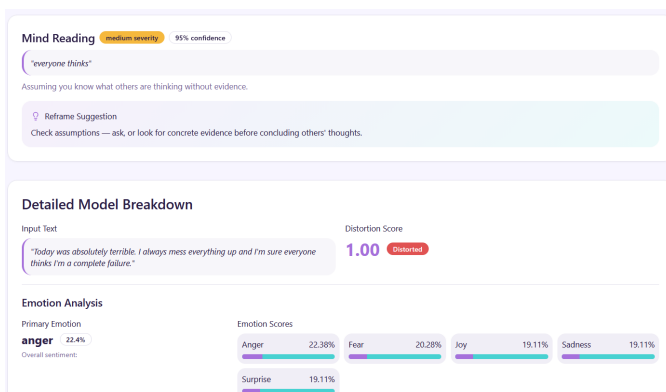


Fig. 3. Detailed Breakdown

C. Screenshots

VIII. PERSISTENT CHALLENGES AND FUTURE DIRECTIONS

A number of enduring issues are revealed by the examination of the field's models and benchmarks. Solving these fundamental, as opposed to computational, issues is critical to the field's future.

A. Challenge: Conceptual and Taxonomic Standardization

The "fragmentation" of the field is the fundamental issue. Different definitions, labels, and groupings are used in different studies, which makes it practically impossible to compare results and reproduce models.

The field needs to "standardize taxonomy adoption" in the future. Researchers should avoid developing new, custom taxonomies without solid clinical explanation, progress toward a consensus taxonomy (such as the often referenced Burns list), and explicitly mention the source and reasoning behind their taxonomy.

B. Challenge: The "Annotation Ceiling" and Data Quality

Low Inter-Annotator Agreement (IAA), which is directly related to the unclear taxonomies, is known to limit model performance. The research by Tauscher et al. (2022) shows that models have already attained this limit. [1]

Future Direction: Annotation techniques must become the primary focus of study instead of model complexity. To increase the human F1-score ceiling, the profession needs "clearer task definitions" and better annotator training manuals. Model performance can only be significantly improved by raising the caliber and consistency of the "ground truth" data.

C. Challenge: The Overreliance on Short Text (Context-Dependency)

The "vast majority" of current datasets structure the classification job at the phrase or single-utterance level, which is a fundamental methodological shortcoming in the area. This is not feasible. Many distortions are highly "context-dependent" and are only subtly indicated by certain words (like "should"). Models are forced to rely on superficial, surface-level patterns as a result of this "stripping away" of context, which results in subpar performance in unclear, real-world scenarios.

Future Direction: "Make context-aware models the norm" is the goal of future research. In order to capture "multi-turn conversations," "patient narratives," and "longer-form content" that more closely resembles therapeutic language, new datasets must go beyond single phrases.

D. Challenge: Ethical Implications and Real-World Application

Although this technology aims to scale online therapy, provide immediate feedback in automated cognitive behavioral therapy, and notify physicians of client thought patterns, there are serious ethical risks. [4]

The information is "highly sensitive" and frequently comes from social media users who did not provide permission for their data to be utilized for this kind of analysis or from private therapy transcripts. Additionally, there is a significant chance of misdiagnosis when an unregulated algorithm is used in a mental health setting.

While a "false negative" (ignoring a serious distortion) could give false reassurance and discourage a user from seeking assistance, a "false positive" (marking a healthy thought as distorted) could cause worry and iatrogenic injury. Another significant issue is algorithmic bias, which occurs when models trained on the language of one group are unable to comprehend or incorrectly classify the language of another.

Future Direction: Transparent and uniform ethical norms must be adopted by the field. Creating a "standardised datasheet template" and a "ethics flowchart" are essential first stages. These tools would increase accountability and safety by mandating that researchers formally record data "provenance, consent, and documentation before experimentation".

CONCLUSION

The categorization of cognitive distortions has been traced in this review from its origins in Cognitive Behavioral Therapy by Beck and Burns to a challenging, high-stakes computer problem. From feature-based SVMs to the present state-of-the-art transformer models, the discipline has experienced a rapid methodological change.

This review's main argument is that the science of cognitive distortion classification is at a turning point. Instead of being model-limited, it is data-limited and definition-limited. The main obstacles are now basic and multidisciplinary rather than computational. These bottlenecks consist of:

- **Taxonomic Fragmentation:** "Fragmented" and non-comparable research results from the absence of a uniform, clinically informed taxonomy.
- **The "Annotation Ceiling":** Our best models have already reached the low ceiling for human-annotator agreement caused by the taxonomies' ambiguity. [1]
- **De-contextualized Datasets:** Models are trained on an overly simplistic task that does not represent real-world, context-dependent clinical language due to an excessive dependence on short-text, single-utterance data. [1]

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